

PRAKTIKUM SISTEM OPERASI

MODUL VI

PROSES SEKUENSIAL DAN KONKUREN



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

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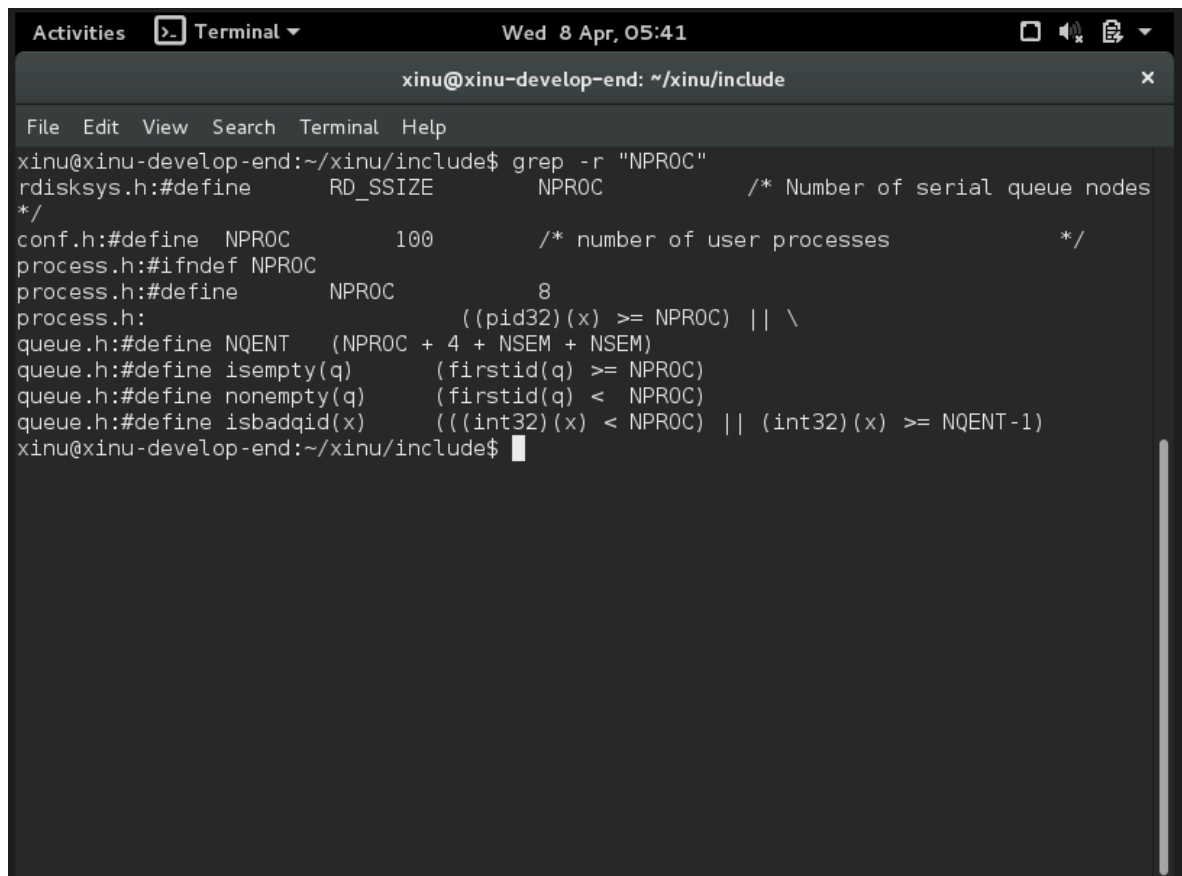
2026

JURNAL

1. Soal 1

Selain hardware (memory), batasan maksimal proses dapat ditentukan dengan secara software. Pada Linux maksimal proses adalah 4194303 proses (64 bit) dan 32767 proses (32 bit) dapat dilihat melalui perintah `$cat /proc/sys/kernel/pid_max`. Carilah pada source code Xinu yang memberi batasan mengenai banyaknya proses yang bisa dibuat! Berapa maksimal proses dalam Xinu? Ubah menjadi maksimal 150 proses!

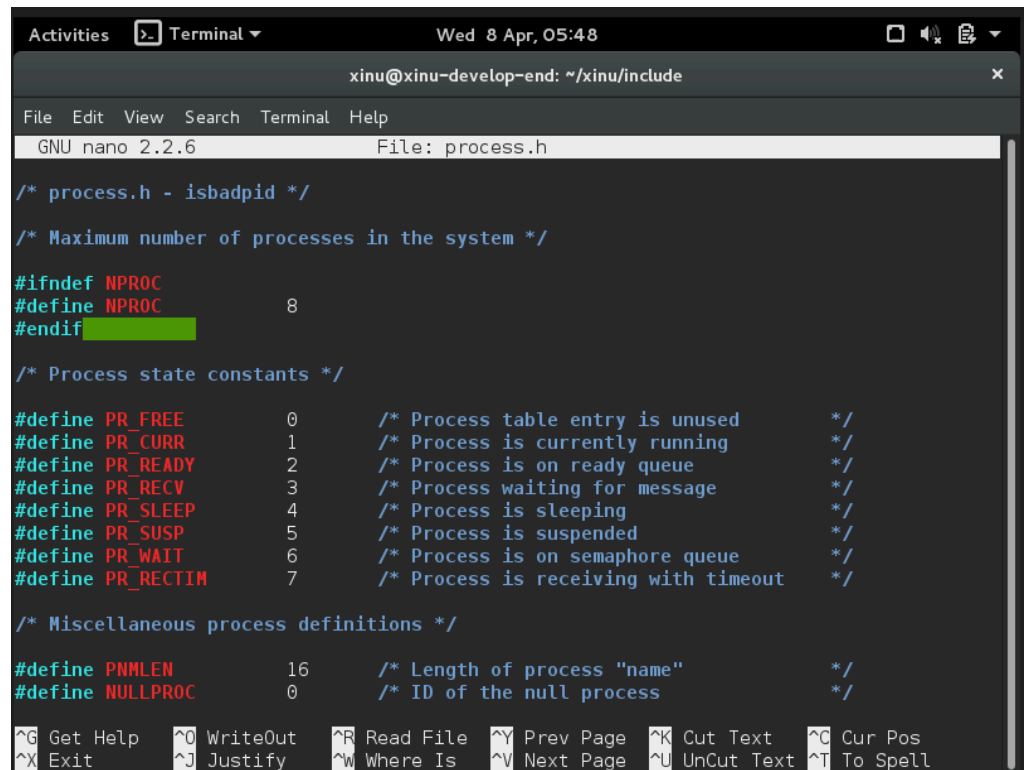
- Menemukan file untuk konfigurasi yaitu berada di direktori 'xinu/include' kemudian di file process.h



```
Activities Terminal Wed 8 Apr, 05:41
xinu@xinu-develop-end: ~/xinu/include

File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu/include$ grep -r "NPROC"
rdisksys.h:#define RD_SSIZE NPROC /* Number of serial queue nodes */
conf.h:#define NPROC 100 /* number of user processes */
process.h:#ifndef NPROC
process.h:#define NPROC 8
process.h: ((pid32)(x) >= NPROC) || \
queue.h:#define NQENT (NPROC + 4 + NSEM + NSEM)
queue.h:#define isempty(q) (firstid(q) >= NPROC)
queue.h:#define nonempty(q) (firstid(q) < NPROC)
queue.h:#define isbadqid(x) (((int32)(x) < NPROC) || (int32)(x) >= NQENT-1)
xinu@xinu-develop-end:~/xinu/include$
```

- Kemudian gunakan 'nano' untuk mengubah file pada process.h untuk mengubah nilai proses menjadi 150
 - Before



A terminal window titled 'xinu@xinu-develop-end: ~/xinu/include' showing the nano 2.2.6 editor editing 'process.h'. The file content is as follows:

```
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */

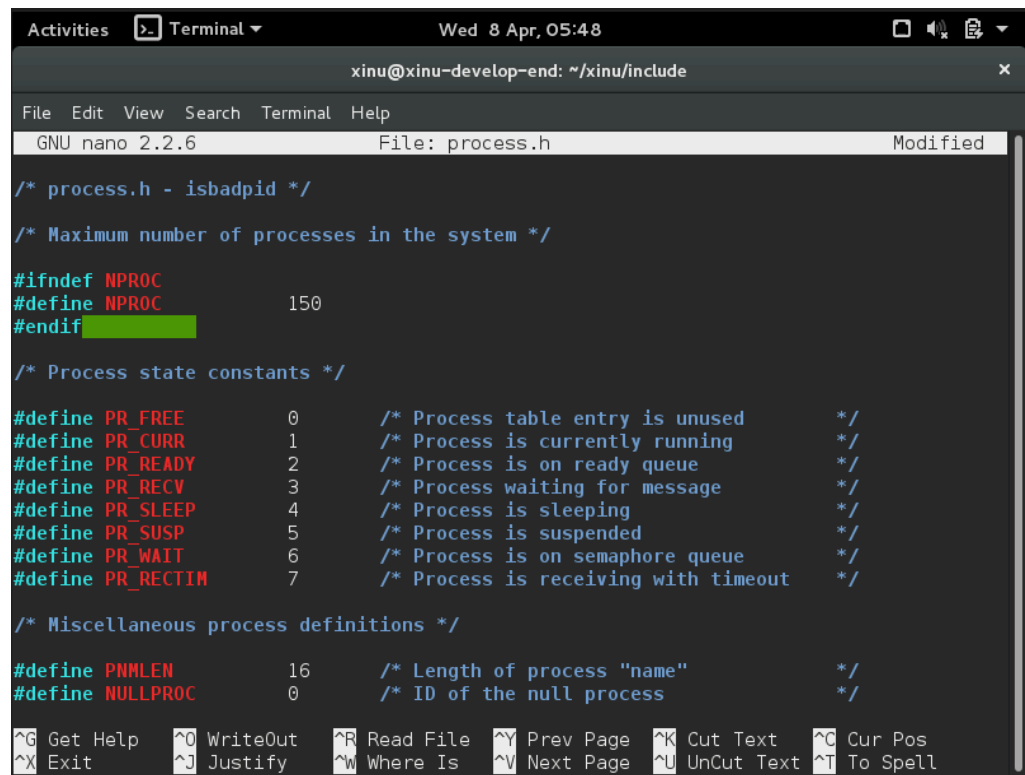
#define PR_FREE    0      /* Process table entry is unused      */
#define PR_CURR    1      /* Process is currently running        */
#define PR_READY   2      /* Process is on ready queue           */
#define PR_RECV    3      /* Process waiting for message         */
#define PR_SLEEP   4      /* Process is sleeping                 */
#define PR_SUSP    5      /* Process is suspended                */
#define PR_WAIT    6      /* Process is on semaphore queue       */
#define PR_RECTIM  7      /* Process is receiving with timeout   */

/* Miscellaneous process definitions */

#define PNMLEN      16     /* Length of process "name"           */
#define NULLPROC    0     /* ID of the null process              */
```

The bottom status bar shows keyboard shortcuts: ^G Get Help, ^O WriteOut, ^R Read File, ^Y Prev Page, ^K Cut Text, ^C Cur Pos, ^X Exit, ^J Justify, ^W Where Is, ^V Next Page, ^U UnCut Text, ^T To Spell.

- After



The same terminal window shows the file 'process.h' after modification. The value of NPROC has been changed from 8 to 150. The status bar now includes an additional 'Modified' indicator on the right.

```
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      150
#endif

/* Process state constants */

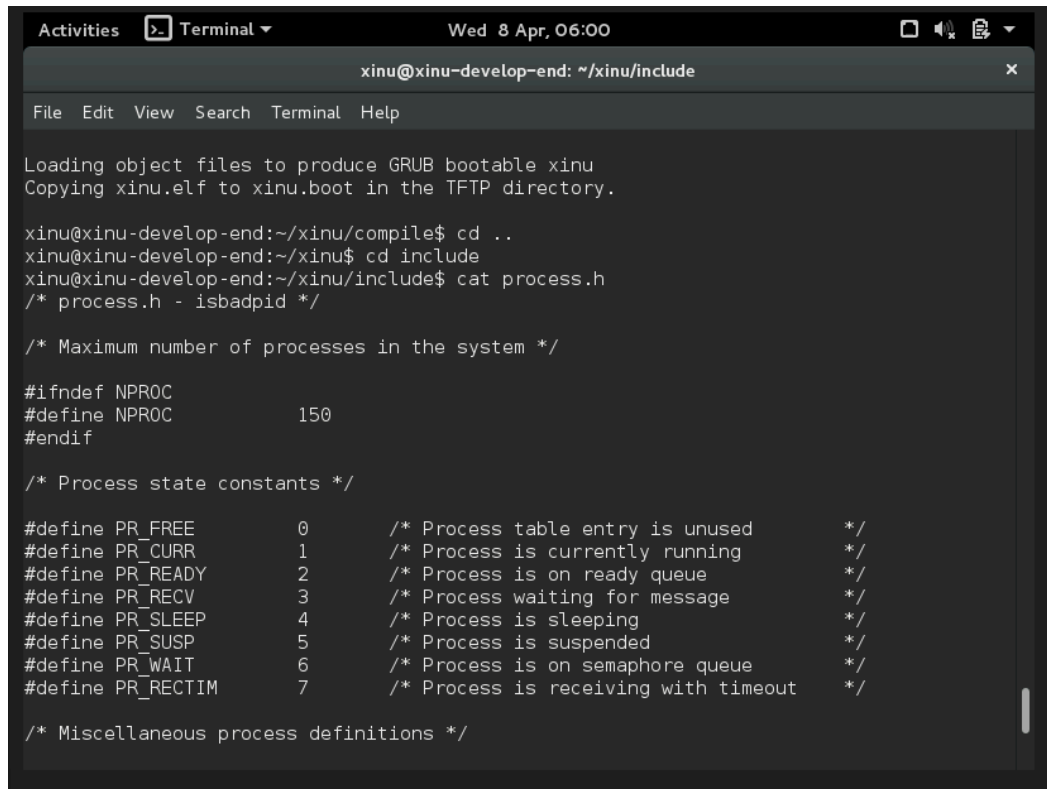
#define PR_FREE    0      /* Process table entry is unused      */
#define PR_CURR    1      /* Process is currently running        */
#define PR_READY   2      /* Process is on ready queue           */
#define PR_RECV    3      /* Process waiting for message         */
#define PR_SLEEP   4      /* Process is sleeping                 */
#define PR_SUSP    5      /* Process is suspended                */
#define PR_WAIT    6      /* Process is on semaphore queue       */
#define PR_RECTIM  7      /* Process is receiving with timeout   */

/* Miscellaneous process definitions */

#define PNMLEN      16     /* Length of process "name"           */
#define NULLPROC    0     /* ID of the null process              */
```

The bottom status bar shows the same keyboard shortcuts as before, plus the 'Modified' status indicator.

c. Berikut setelah dilakukan Perubahan pada process.h



A terminal window titled 'xinu@xinu-develop-end: ~/xinu/include' showing the following commands and output:

```
File Edit View Search Terminal Help

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$ cd ..
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ cat process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC      150
#endif

/* Process state constants */

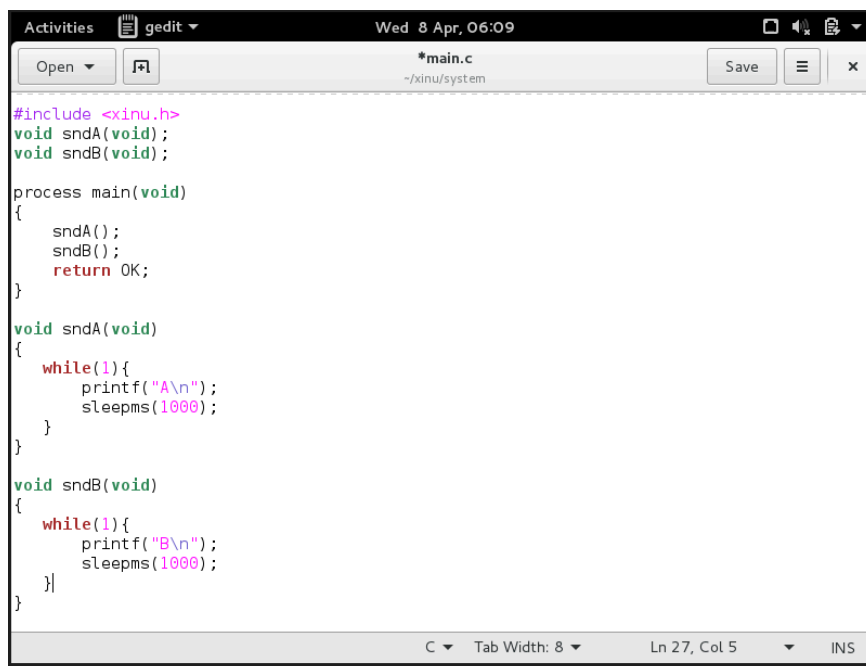
#define PR_FREE      0      /* Process table entry is unused      */
#define PR_CURR      1      /* Process is currently running      */
#define PR_READY     2      /* Process is on ready queue      */
#define PR_RECV      3      /* Process waiting for message      */
#define PR_SLEEP     4      /* Process is sleeping      */
#define PR_SUSP      5      /* Process is suspended      */
#define PR_WAIT      6      /* Process is on semaphore queue      */
#define PR_RECTIM    7      /* Process is receiving with timeout */

/* Miscellaneous process definitions */
```

2. Soal 2

Jalankan kode main_ex1!

a. main_ex1 syntax:



A gedit editor window titled 'xinu@xinu-develop-end: ~/xinu/system' showing the following C code:

```
*main.c
~/xinu/system

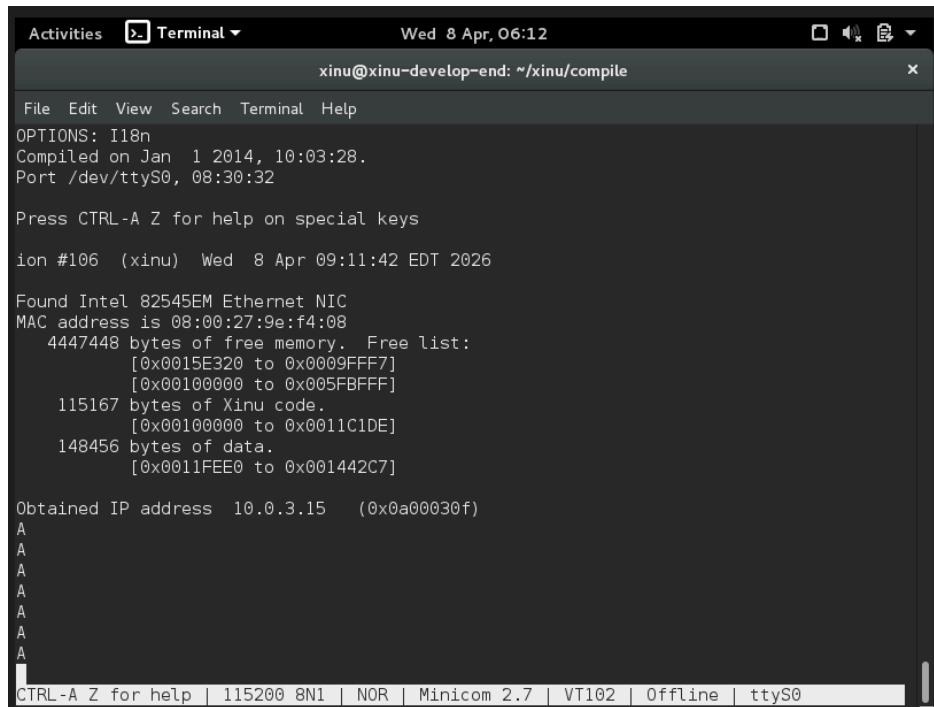
#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    sndA();
    sndB();
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}
```

b. hasil running



```
Activities Terminal Wed 8 Apr, 06:12
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
OPTIONS: I18n
Compiled on Jan 1 2014, 10:03:28.
Port /dev/ttyS0, 08:30:32

Press CTRL-A Z for help on special keys

ion #106 (xinu) Wed 8 Apr 09:11:42 EDT 2026

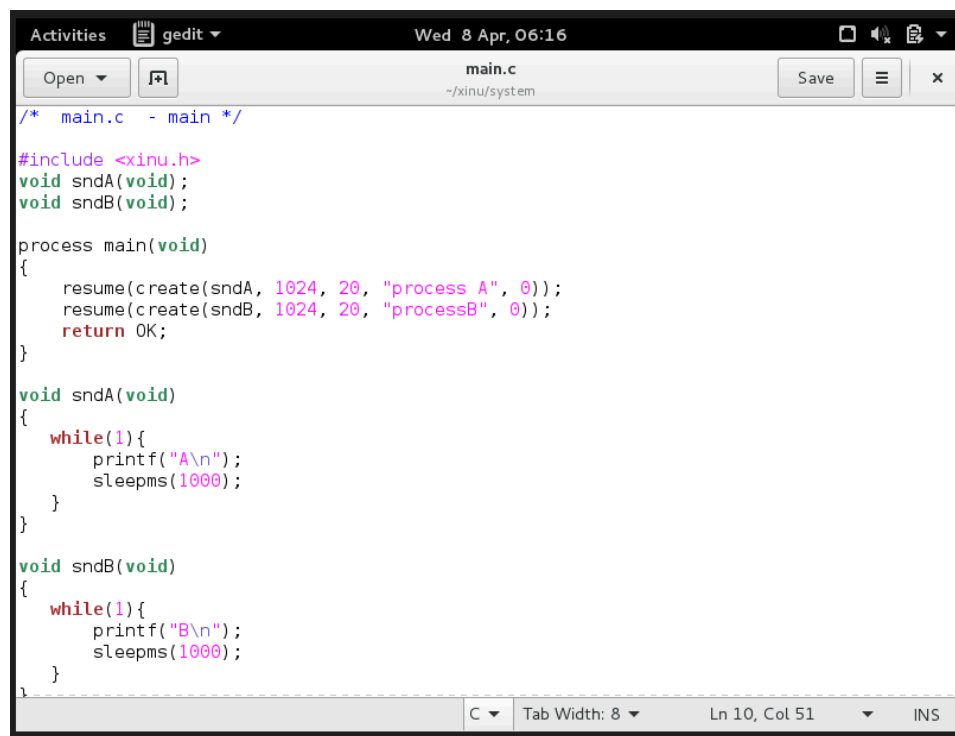
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFFF]
    [0x00100000 to 0x005FBFFF]
  115167 bytes of Xinu code.
    [0x00100000 to 0x0011C1DE]
  148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
A
A
A
A
A
A
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

3. Soal 3

Jalankan kode main_ex2!

a. main_ex2 syntax:



```
Activities gedit Wed 8 Apr, 06:16
main.c
~/xinu/system

Open Save
/* main.c - main */

#include <xinu.h>
void sndA(void);
void sndB(void);

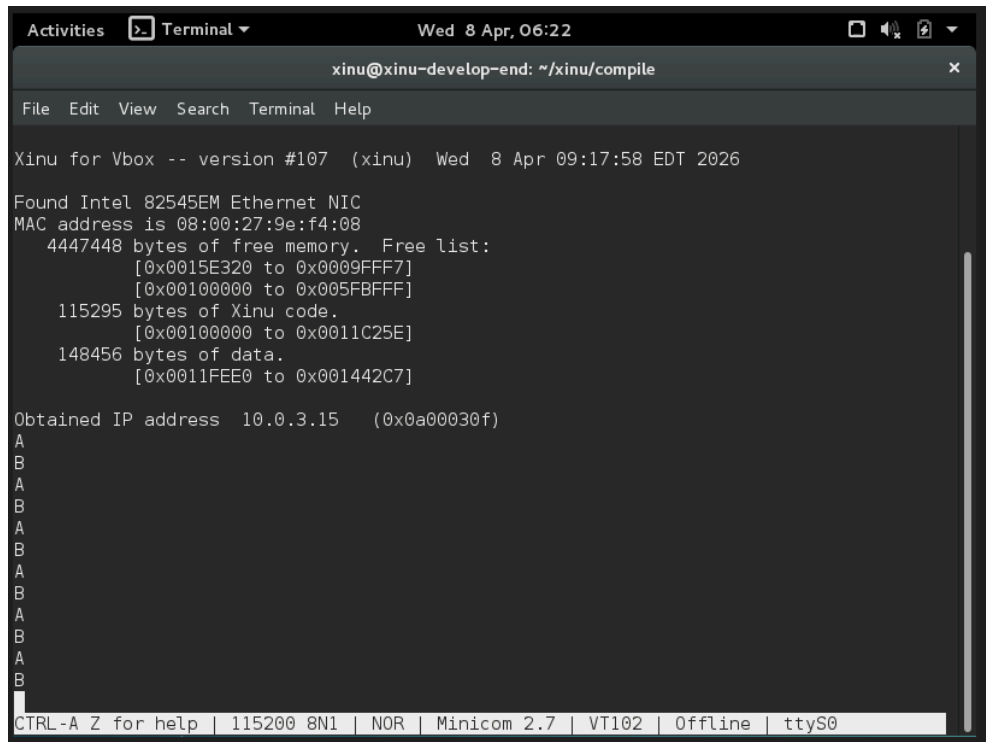
process main(void)
{
    resume(create(sndA, 1024, 20, "process A", 0));
    resume(create(sndB, 1024, 20, "processB", 0));
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}

C Tab Width: 8 Ln 10, Col 51 INS
```

b. hasil running:



A terminal window titled 'xinu@xinu-develop-end: ~/xinu/compile' showing the output of the Xinu boot process. The output includes the version number (#107), the date and time (Wed 8 Apr 09:17:58 EDT 2026), the discovery of an Intel 82545EM Ethernet NIC with MAC address 08:00:27:9e:f4:08, memory statistics (4447448 bytes of free memory, 115295 bytes of Xinu code, 148456 bytes of data), and the obtained IP address (10.0.3.15). The output also shows a series of 'A' and 'B' characters being sent and received, and a status bar at the bottom indicating 'CTRL-A Z for help' and other system information.

```
Activities Terminal Wed 8 Apr, 06:22
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

Xinu for Vbox -- version #107 (xinu) Wed 8 Apr 09:17:58 EDT 2026

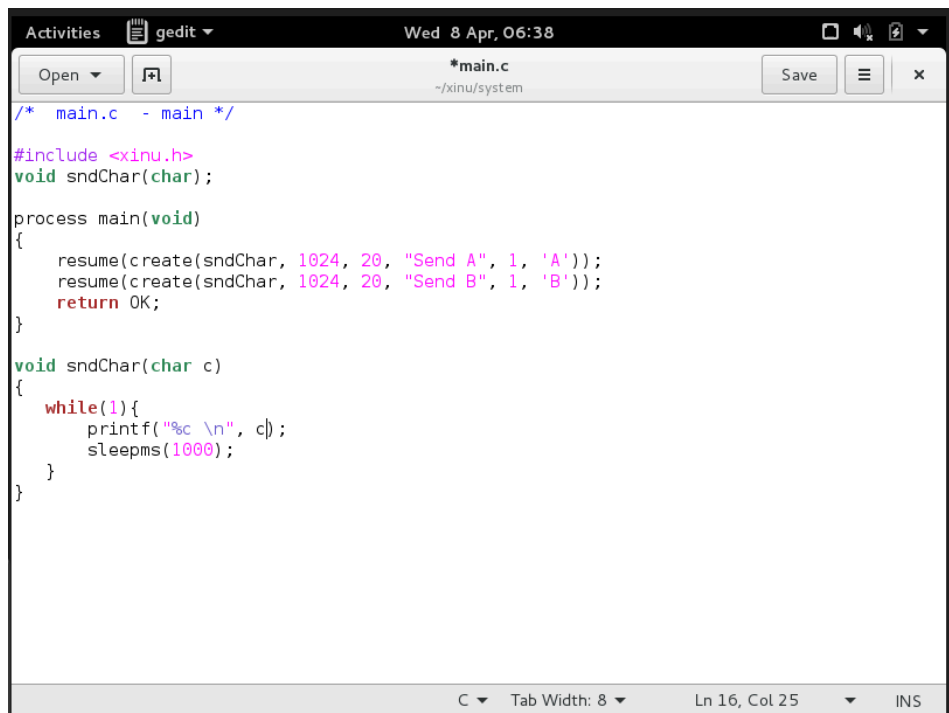
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
  115295 bytes of Xinu code.
    [0x00100000 to 0x0011C25E]
  148456 bytes of data.
    [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
A
B
A
B
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

4. Soal 4

Jalankan kode main_ex3!

a. main_ex3 syntax:



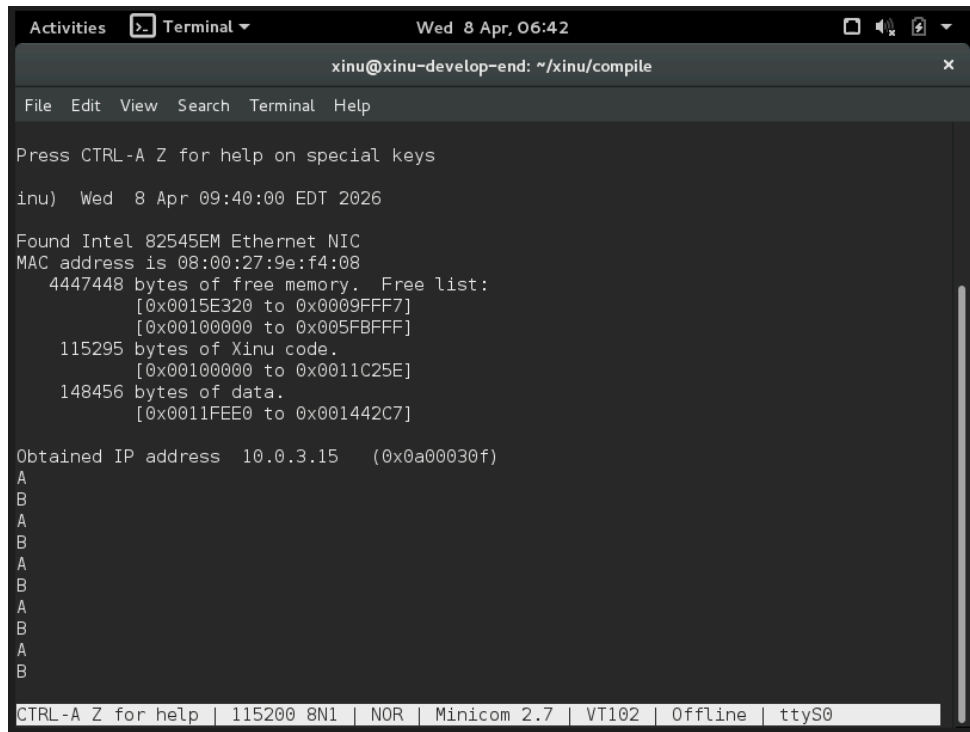
A Gedit editor window titled 'gedit' showing the source code for 'main.c'. The code includes a header file 'xinu.h', defines a 'sndChar' function, and implements a 'main' function that creates two processes, 'Send A' and 'Send B', and returns 'OK'. The 'sndChar' function is a loop that prints the character and sleeps for 1000ms.

```
Activities gedit Wed 8 Apr, 06:38
*main.c
~/xinu/system
Open Save
/* main.c - main */
#include <xinu.h>
void sndChar(char);

process main(void)
{
    resume(create(sndChar, 1024, 20, "Send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "Send B", 1, 'B'));
    return OK;
}

void sndChar(char c)
{
    while(1){
        printf("%c \n", c);
        sleepms(1000);
    }
}
```

b. hasil running:



```
Activities Terminal Wed 8 Apr, 06:42
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

Press CTRL-A Z for help on special keys

inu) Wed 8 Apr 09:40:00 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
  [0x0015E320 to 0x0009FFF7]
  [0x00100000 to 0x005FBFFF]
115295 bytes of Xinu code.
  [0x00100000 to 0x0011C25E]
148456 bytes of data.
  [0x0011FEE0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
A
B

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

5. Soal 5

Buatlah 2 proses produser dan konsumen. Produser memproduksi angka integer dari 1-1000. Konsumer mengkonsumsi integer yang diproduksi oleh produser dan menampilkannya! (Gunakan variabel global bertipe int32 bernama n yang digunakan secara bersama oleh kedua proses)

```
int32 n = 0; //global variabel

void produser(void){
    int32 i;
    for (i=1; i<=1000; i++){
        n++;
    }
}

void konsumer(void){
    int32 i;
    for (i=1; i<=1000; i++){
        printf("Nilai dari n adalah %d\n",n);
    }
}
```

a. syntax yang digunakan:\

b. hasil outputnya:

A screenshot of a Linux terminal window titled "Terminal". The window has a dark theme. At the top, there's a title bar with "Activities", a terminal icon, "Terminal", and system information "Wed 8 Apr, 06:59". Below the title bar is a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The main area shows the prompt "xinu@xinu-develop-end: ~/xinu/compiler" followed by 20 lines of output, each reading "Nilai dari n adalah 1000". A vertical scrollbar is visible on the right side of the terminal pane.

- c. Hasil dari program produser-konsumer surprise karena adanya kondisi balapan. Operasi $n++$ tidak bersifat atomik (terdiri dari tiga instruksi: LOAD, ADD, STORE). Sistem operasi dapat melakukan pengalihan konteks di antara instruksi-instruksi tersebut, sehingga nilai yang dibaca oleh konsumen menjadi tidak berurutan dan tidak dapat diprediksi. Solusinya adalah dengan menggunakan semaphore untuk melakukan sinkronisasi.